HW4- Computer Simulation II

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1

We can calculate the autocorrelation time using the magnetization, plotting it in a log scale we get:

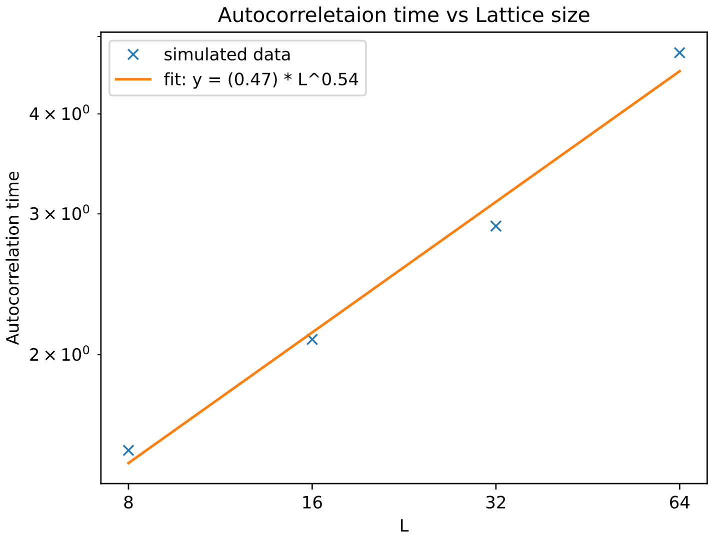

Figure 1: Autocorrelation time for the clusters

To convert this into units of lattice sweeps we use:

$$\tau_{\text{sweep}} = \tau_{\text{cluster}} \cdot \frac{\langle s \rangle}{L^2}. \tag{1}$$

Where $\langle s \rangle$ is the average cluster size. Plotting in log scale we get:

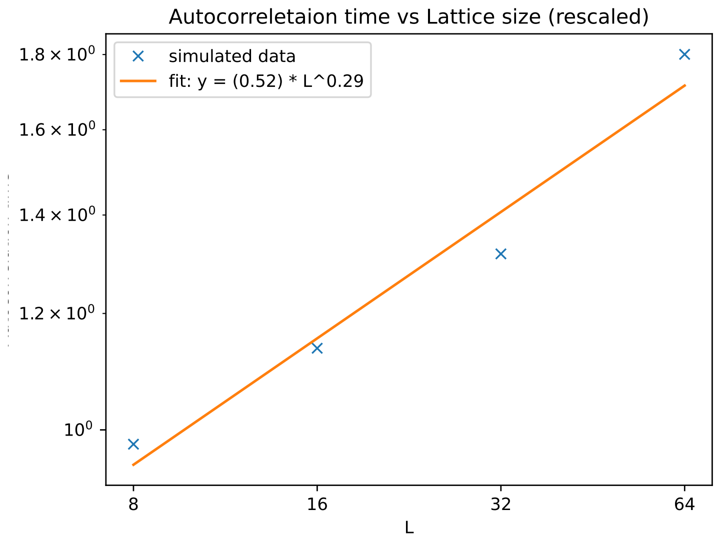

Figure 2: Rescaled correlation time

So we can see:

$$\tau \propto L^{0.29}$$

This is significantly less then the autocorrelation time of the Metropolis algorithm $\tau \propto L^{2.17}$, but still a bit larger then the expected amount for the Wolff algorithm $z \approx 0.13$.

2

Calculating the value of χ we can clearly see the peaks around the critical temperature:

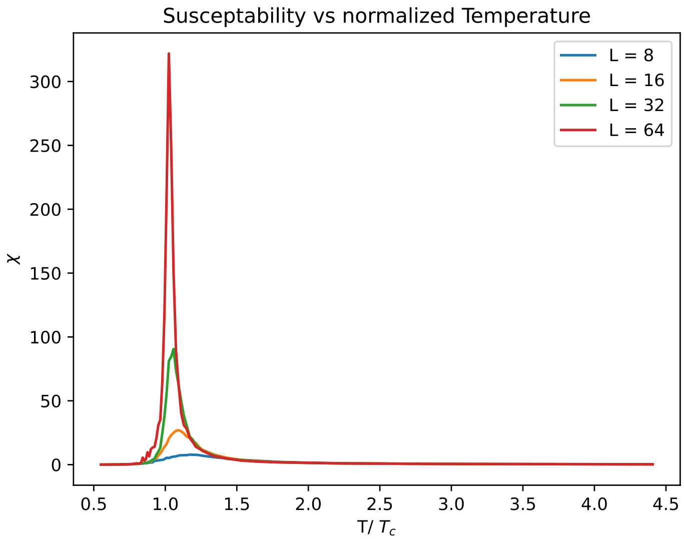

Figure 3: χ values

Taking the maximum of χ and plotting it against the lattice size L, we can see the relation between them scales by a power law:

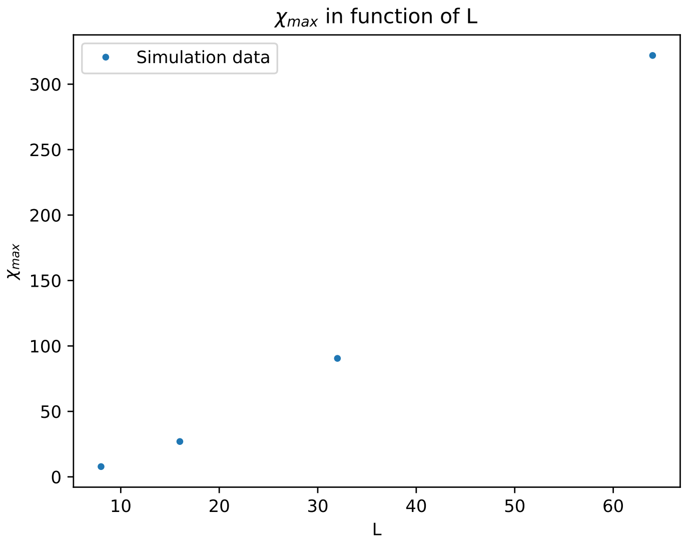

Figure 4: χ in function of L

For us to test the finite-size scaling Ansatz, we first need to make some adjustments, instead of using the given formula we shall work in log space:

$$ln(\chi) = \frac{\gamma}{\nu} ln(L) + c \tag{2}$$

Where c is a proportionality constant. Using this equation and fitting a line we can find the values for γ/ν and c.

$$\frac{\gamma}{\nu} = 1.783169... \quad c = -1.65374 \tag{3}$$

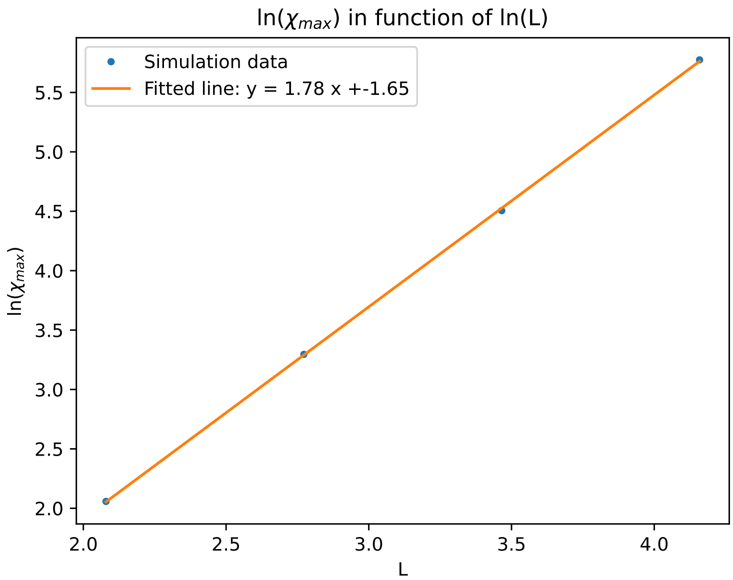

Figure 5: Fitted χ_{max}